

A Technical Comparison of the Studer A80 Tape Machine and the Studer A800 Plug-In

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Abstract

Are modern digital tape emulation plugins able to replicate the sound of analog tape machines? A comparison of a hardware analog tape machine and a tape emulation plugin was performed to analyze common characteristics with the expectation that there would be no significant differences between them. Tests were performed using a late 1970s-era Studer A80 RC MkII ¼" two-track analog recorder and the Universal Audio Designs Studer A800 Multichannel Tape Recorder v1.10 analog tape machine emulation plugin. This plugin was selected as the electronics modeled for this plugin most closely match the selected hardware machine. The hardware machine and the plugin were calibrated to a level of 520 nanowebers per meter (+9) using the same tape formulation. An Audio Precision Audio Precision SYS 2522 System Two Cascade Dual Domain analyzer was used to measure Frequency Response, Signal to Noise Ratio (SNR), and Total Harmonic Distortion (THD) at different machine settings and tape speeds (Input, 15 IPS reproduce playback, and 30 IPS reproduce playback). Results showed that the Studer A800 plugin reproduced the frequency response of the Studer A80 without significant differences, with the exception of the low range at 30 IPS. Minor differences in SNR and THD were noted.

Introduction

1.1 Analog Recording vs Digital Emulation, a Statement of the Problem

Going back to the earliest days of music recording and production, engineers in the field of audio have always sought to adopt technological advancements wherever possible offering superior sound quality or aiding in streamlining creative workflows both cost wise as well as for ease of workflow. Some of these innovations include going from mono to stereo to Dolby Atmos (spatial audio); from direct two-track to multitrack recording; and from analog music recording and mixing to performing these tasks in the box. Naturally, as these innovations have occurred, it has often led to a variety of opinions of engineers within the industry when it comes to what way to record is the "best". These differences can often be displayed generationally, where engineers who have done it for twenty-thirty years will tend to prefer to do what they have the most experience doing (using analog techniques), while younger engineers will tend towards mixing in the box for ease while not having a full understanding of older mixing flows.

It is the responsibility of the industry as a whole to design technology with plugins to try to bridge this gap by attempting to provide the same tools and sounds to engineers that have historically been captured from the analog world into the digital world. One of the most significant areas for this is with tape emulation plugins. Analog tape machines and tape has continued to get more and more expensive and more and more difficult for the average engineer to be able to afford and use in their workflows. Tape emulation plugins are an important way to help provide wide access to this same technology in a more cost-efficient way.

With this said, the accuracy of these emulations needs to be as high as possible to be widely adopted by engineers in this industry. There are a variety of opinions that have been presented on this accuracy, both from plugin companies that market their plugins to say they are providing super accurate reproductions (1), to audio engineers who have yet to be fully satisfied with the digital reproduction in their creative workflows (2).

1.2 Research Question

Does the tape emulation plugin provide an accurate and faithful representation of the hardware machine in terms of Frequency Response, SNR, and THD at the corresponding settings?

Methods

2.1 Analog Experimental Design

An XLR cable was connected from the measuring laptop equipment running Audio Precision to the input of the machine. The output of the machine was connected to the output of the measuring laptop, where the corresponding measurements were taken. The reproduce head of the machine was calibrated at 15 IPS and 30 IPS using an MRL Calibration Tape (250 nWb/m) with test tones of 100 Hz, 1kHz, and 10kHz. Sine wave test signal from the Audio Precision was used to align the record head of the machine for a level of 520 nWb/m for 15 IPS and 30IPS.

2.2 Tape Type Used

A new reel of Emtec Studio Master 900 maxima ¼" tape was used for this project. This tape was selected for compatibility with the plugin, which included it as a selection.

2.3 Digital Experimental Design

The Studer A800 plugin was loaded onto an empty track on Pro Tools. An XLR cable was then run from the measuring laptop equipment running Audio Precision to the input of tie line 13 and 14. The output of the track on Pro Tools was then run back to the output of the measuring laptop and the corresponding measurements were taken. The studio was patched to where tie lines 13 and 14 were made to channels 1 + 2 on the Audio Interface. Proper signal flow and levels were tested and seen to be working correctly before the measurements were ultimately taken.

2.4 Tape Selection, Audio Interface Used and Misc. Plug-In Settings Used

The Studio Master 900 maxima tape selection was selected for this project (same as the one used for Analog).

This test was run through an Apollo x8p (48 kHz Sample Rate, 24 Bit Depth, 1.1 milliseconds of latency @ 96 kHz Sample Rate) on Line Inputs 1 and 2 (Frequency Response 20 Hz-20kHz, +/- 0.04 dB, 123 Dynamic Range (A-weighted), -113 dB THD + N, 24 dBu Maximum Input Level, 10K Ohms Input Impedance, +10 dB to +65 dB Gain Range).

The test was performed with the NAB EQ Setting, the Noise setting On, and the Auto Cal button pressed on. The A800 plugin would be recalibrated each time going from each of the variety of settings to ensure most proper calibration and results.

2.5 Experimental Design

Sine Sweeps were run through the Tape Machine in Input Mode, and in Record Mode at 15 IPS and 30 IPS tape speeds. After the Frequency Responses were captured with the Sine Sweep at a chosen mode (i.e. input mode), the THD+N and SNR measurements were simultaneously recorded at this setting before moving to the next setting. After

the Tape Machine measurements were done, patches were then shifted to perform the same measurements on the A800 plugin.

Pictures of Digital Experimental Design Setups

Figure 1: 30 IPS



Figure 2: 15 IPS



Results

3.1 Tape Machine Results.... Analysis by itself

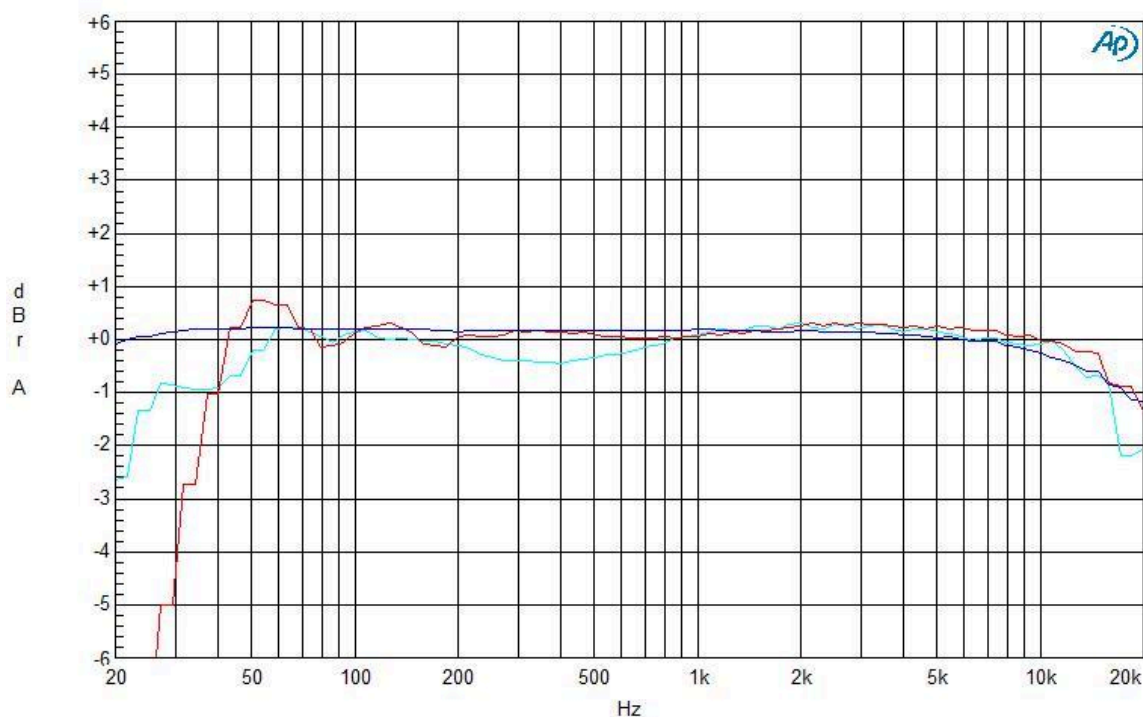
At Input mode, the Frequency Response captured was found to be relatively flat (Figure 1) with a SNR of -68.85 dBrA and a THD+N Ratio of 0.08%.

At Record mode at 15 IPS, the Frequency Response was captured as shown in Figure 1, with a SNR of -60.3 dBrA and a THD+N Ratio of 0.26%.

At Record Mode at 30 IPS, the Frequency Response was captured as shown in Figure 2, with a SNR of -62.5 and a THD+N ratio of 0.20%.

Figure 3: Hardware Input vs 15 IPS vs 30 IPS

Audio Precision Frequency Response Sine Sweep +4 Overbias 07/10/24 16:44:05
Studio Master 900 Tape



Sweep	Trace	Color	Line Style	Thick	Data	Axis	Comment
1	1	Cyan	Solid	1	Anlr.Level A	Left	Amp.;15 IPS,+9bias, 1K
2	1	Red	Solid	1	Anlr.Level A	Left	Amp. ; 30 IPS, +9Bias, 1K
30	1	Blue	Solid	1	Anlr.Level A	Left	Machine Input

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3.2 Plugin Emulation Results... Analysis by Itself

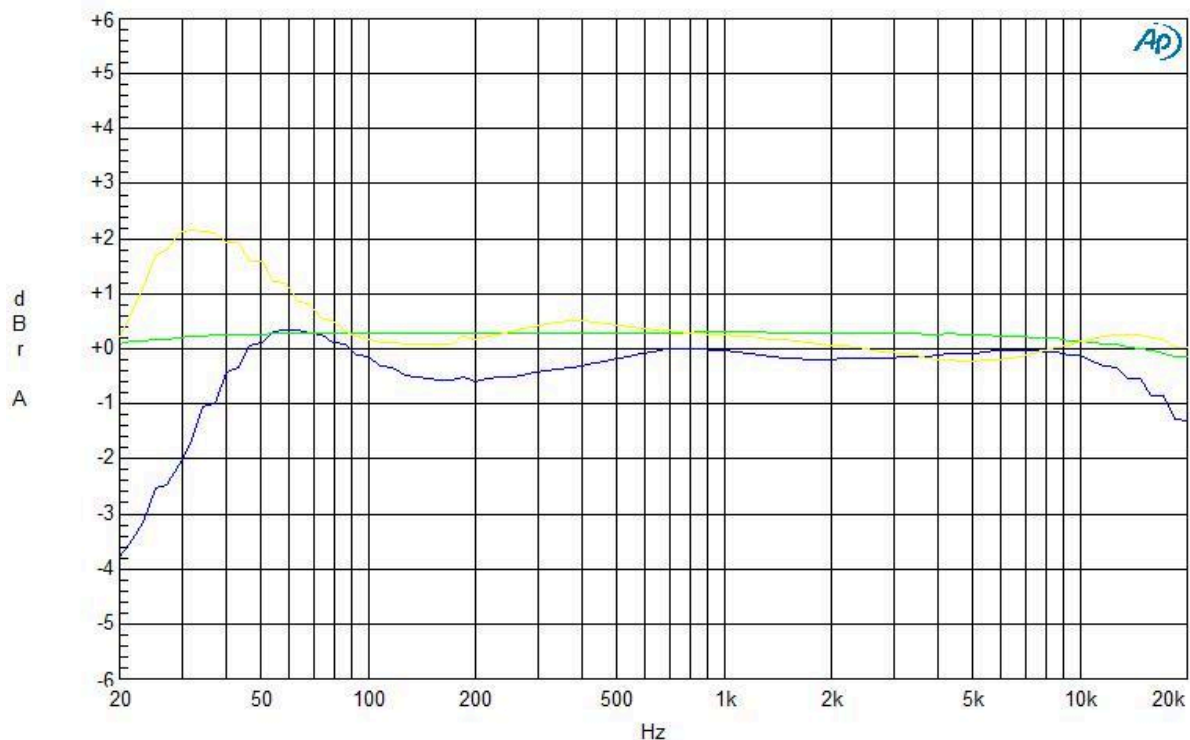
At input mode, the Frequency Response captured was found to be relatively flat(Figure 2) with a SNR of -68.1 dBrA and a THD +N Ratio of .074%.

At Record mode at 15 IPS, the Frequency Response was captured as shown in Figure 1, with a SNR of -67.8 dBrA and a THD+N Ratio of 0.26%.

At Record Mode at 30 IPS, the Frequency Response was captured as shown in Figure 2, with a SNR of -67 and a THD+N ratio of 0.54%.

Figure 4: Digital Emulation Input vs 15 IPS vs 30 IPS

Audio Precision Plugin Frequency Response at Input, 15 IPS, and 30 07/12/24 10:51:42
IPS



Sweep	Trace	Color	Line Style	Thick	Data	Axis	Comment
8	1	Blue	Solid	1	Anlr.Level A	Left	Plugin Amp; 30 IPS, +9 Bias
9	1	Green	Solid	1	Anlr.Level A	Left	Plugin Amp; Input
10	1	Yellow	Solid	1	Anlr.Level A	Left	Plugin Amp; 15 IPS, +9 Bias

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Discussion

Observations of Frequency Response Data

Our findings of the data in this category showcased a pretty accurate emulation between Digital and Hardware for the most part.

In Input mode (input lines shown in the graphs on Figures 3 and 4), the Frequency Response graphs of the Hardware vs. the Plugin showed to be nearly identical to each other. The only difference was there was a cut in the high end to about 1 dBrA in the high end in the hardware that was not captured by the Plugin (from about 8 kHz to 20 kHz). While this difference may be slightly significant soundwise to the most strict engineer, the manufacturer's goal of the equipment in Input mode is for a perfectly flat response in input (input is not a recording mode, hence you aren't losing anything with a difference in the performance), so the digital plugin ends up capturing this more accurately. Even though this difference was captured, this difference isn't significant enough in any way since these two graphs are completely identical in functional reality. However, if you are an engineer that is trying to emulate what you would hear out of just the Tape Machine speaker, one could add an EQ to scoop out some of the highs in that range as shown in the graph for the Input in Figure 3. It's probably not something any normal engineer would look to do anything about due to the actual function of what Input Mode is supposed to accomplish in reality.

In Record Mode at 30 IPS, the Frequency Response graphs were shown to be nearly identical to each other within 0.5 dB. The most significant differences in Frequency response were around the 100 to 500 Hz range (the Plugin cut here by 0.5 dB while the Hardware stayed relatively flat), and from about 1-8kHz (Hardware boosted here by 0.25 dB whereas the Plugin stayed perfectly flat).

In Record Mode at 15 IPS, the Frequency Response Graphs were shown to have a pretty significant difference (Figure 6). There was a major bump in the low end from 0-100 Hz that was captured by almost 2 dB in the plugin, whereas the Hardware measurement was cut by almost 3 dB. There were also very slight differences in the 200 to 1 kHz ranges (Plugin boosted by 0.5dB and Hardware cut at 0.5 dB) and from the 2-10kHz ranges (Hardware slightly boosted here by 0.25 dB whereas the Plugin stayed flat).

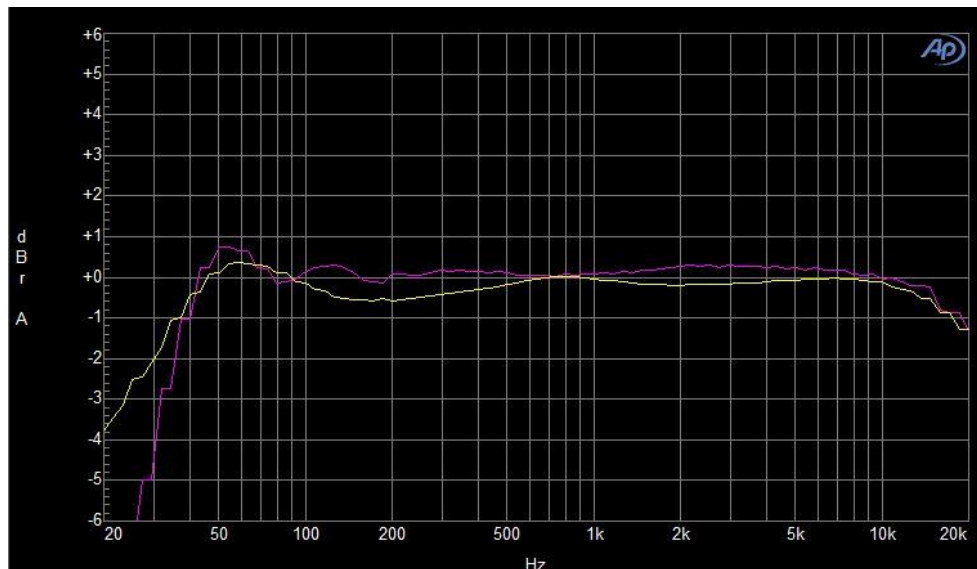
Observations of SNR and THD+N Data

In Input mode, the values for SNR and THD+N between Hardware(-68.85 dBrA SNR, 0.08% THD+N) and the Plugin(-68.1 dBrA SNR, 0.074) were found to be nearly identical.

In Record Mode at 30 IPS, there was shown to be a significant difference in THD+N between Hardware and Plugin (0.20% vs 0.54%) and a slight difference in SNR between Hardware and Plugin (-62.5 dBrA and -67 dBrA). This difference in Harmonic Distortion here is a major problem and error in the comparison. While the SNR's are slightly different, this difference doesn't really seem to be that consequential in the long run.

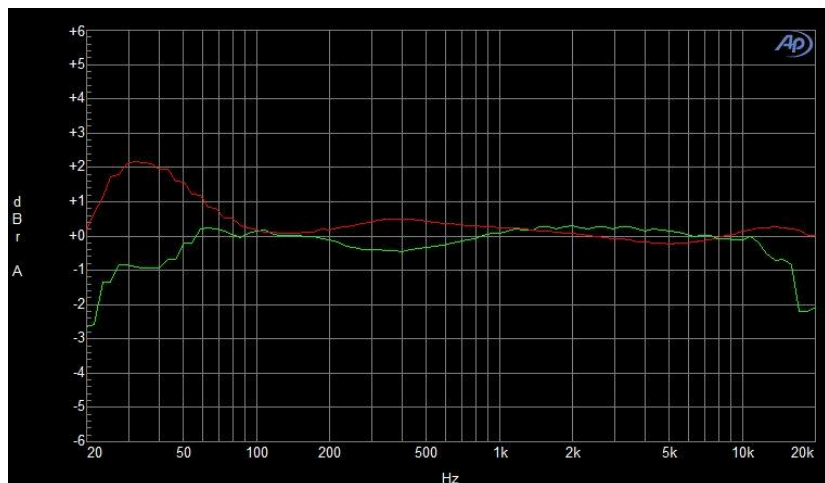
In Record Mode at 15 IPS, the values of THD+N captured in Hardware and Plugin were identical (0.26% and 0.26%). There was a difference in SNR ratio between Hardware and Plugin (-60.3 dBrA and -67.8 dBrA).

Figure 5: Digital Emulation vs. Hardware at 30 IPS



Yellow-Digital
Purple-Hardware

Figure 6: Digital Emulation vs Hardware at 15 IPS



Red-Digital

Green-Hardware

Conclusions

The most significant difference that came up with the results in this experiment was the boost in the low end at 15 IPS on the Plugin. This difference was not as a result of a calibration error (calibration was checked and double checked multiple times), and seems to be a significant result and concern to address/look into. The slight differences found in SNR's, suggest that the plugin keeps a largely consistent SNR in all settings, which isn't going to happen in Hardware Tape Machines. While this might be an issue for some, recording with as little of a SNR as possible is most ideal soundwise (-67.8 vs -60.3 at 15 IPS), so this goes to show that while the two machines weren't identical, the digital plugin performs far better in this area mechanically. In terms of THD+N measurements, there was a significant issue and difference at 30 IPS that would make a major difference soundwise, whereas there was no difference at all in any of the other modes. This difference at 30 IPS is pretty significant and an important one to note. Most engineers that would use a Tape Emulation Plugin would probably lean to using this 30 IPS setting, so that's going to make a pretty major difference and problem. The only way you could say there was a user error here would be for a calibration issue, however extensive care was made throughout the process to make sure everything was calibrated exactly the same.

With all of these points in mind, it is important to note that every individual Tape Machine is going to be slightly different. Factors such as age, wear and tear, and

mechanical issues in the machine can cause deviations from machine to machine. All things considered, the A800 Plugin did a pretty decent job (outside of those two major flaws) of emulating the plugin; especially considering the point above.

Future Research Questions to Pursue

In the future, more Tape Machine Emulations could be tested in this same way with other Two-Track Tape Machines. In addition, these results could be paired with a listening test recording actual musical samples having participants judge whether they could detect a difference between the three modes (using the Digitally captured version for Input).

Citations

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- [2] Chambers-Moranz, Ryland; Pras, Amandine; Thomas, Nate; The Generation Gap—Perception and Workflow of Analog vs. Digital Mixing [PDF]; University of Lethbridge, Lethbridge, AB, Canada; School for Advanced Studies in the Social Sciences, Paris, France; Paper 10300; 2019 Available: <https://aes2.org/publications/elibrary-page/?id=20672>
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